

VOCODE

256 Band-Vocoder Plugin

User Manual · v1.2 · 256-band edition

1. Introduction

VOCODE is a classic filter-bank vocoder plug-in built with the JUCE framework. It splits an audio signal into up to **256 frequency bands**, analyses the amplitude envelope of each band from a modulator source, and uses those envelopes to gate the same bands of a carrier signal. The result is the characteristic "talking synthesiser" effect, or without a sidechain a richly self-gated robotic texture.

Unlike modern FFT-based vocoders, VOCODE uses traditional **cascaded IIR biquad band-pass filters** for its analysis and synthesis banks, and with 256 bands the most you'll probably encounter. This gives it a warm, slightly coloured sound characteristic of hardware vocoders from the 1970s and 80s, while allowing individual band-widths to be sculpted in real time via the interactive display, perfect for explorative genres like atmospheric dub techno.

Changelog:

- Version 1.2: State Saving Bugfixes, Added Preset Import / Export and Randomize / Smooth Buttons, Start and End Frequency Sliders, a Normalizer and if you use exactly 133 bands they can be tuned to 12TET from C-1 to C10 with the Chromatic Lock Toggle.
- Version 1.1: Added more colorways, better visual response, and individual band volume editing.
- Version 1.0: Initial Release

2. Signal Routing

2.1 Standard (Sidechain) Mode

In standard mode the plug-in uses **two input buses**:

- **Bus 0 - Carrier:** The source whose timbre will be shaped.
- **Bus 1 - Modulator (sidechain):** The (harmonically rich) signal whose spectral envelope is imprinted on the carrier. A microphone feed is a most common choice, but any signal with time-varying formants works well.

Connect both buses in your host to engage standard vocoder mode. The modulator bus is marked as an optional input in the plug-in bus description, so hosts that do not support multi-input will still load the plug-in.

2.2 Auto-Vocode Mode

When **no sidechain bus is active**, either because the host has not connected a signal to Bus 1, or because the host exposes fewer than two input buses, VOCODE automatically switches to **auto-vocode mode**. In auto-vocode mode the carrier signal is used as its *own* modulator. Each band gates itself by its own envelope: bands with energy open; silent bands close. The net effect is a **spectrally**

focused, "robotic" self-vocoding, the signal is chopped into frequency slices and each slice is amplitude-followed independently. Think of it as a very precise multi-band gate with an organic attack/release character.

Note The AUTO-VOCODE indicator in the top-right corner of the display lights up whenever the plug-in is running in auto-vocode mode. No action is required, the switch is fully automatic.

Useful things to feed into auto-vocode mode:

- A dense pad or string ensemble - bands with harmonic content stay open, gaps close, giving a gating/tremolo texture that tracks the harmonic rhythm of the source.
- A drum loop - the transient energy opens bands briefly, producing a percussive flutter on whatever is playing.
- A voice - gives a subtle telephony / talkbox timbre without a separate carrier.

3. Controls

3.1 Knob Row

Seven rotary knobs sit across the top of the plug-in window. All knobs respond to vertical mouse drag. Double-click to type a value. Right-click for the host automation menu.

PARAMETER	RANGE	DEFAULT	DESCRIPTION
BANDS	1 – 256	16	Number of frequency bands. Bands are distributed logarithmically from 80 Hz to 12 kHz. Higher counts give finer spectral resolution but increase CPU cost linearly.
ATTACK	1 – 500 ms	10 ms	Envelope follower attack time, per band. Faster attack captures transients more tightly but can introduce audible AM artefacts at low values (< 3 ms).
RELEASE	10 – 2000 ms	120 ms	Envelope follower release time, per band. Longer release smooths the gating effect; shorter release gives a more staccato, chattering character.
BAND ORDER	1 – 4	1	Number of cascaded biquad stages per band. Higher order = steeper band-pass skirts = better channel separation. Each additional stage roughly halves cross-talk at the cost of slightly increased latency and CPU.
COMP	0 – 1	0	Post-vocoder compressor amount. At 0 the compressor is bypassed. At 1 the ratio is 20:1 with a fixed –12 dBFS threshold and 10 dB make-up gain. Useful to tame level variance across different band counts.

PARAMETER	RANGE	DEFAULT	DESCRIPTION
DRY / WET	0 – 1	1 (wet)	Crossfade between the unprocessed carrier (dry) and the vocoded output (wet). At 0.5 both are summed at equal level. The dry signal is the carrier; the modulator is never audible in the dry path.
GAIN	0 – +24 dB	0 dB	Output gain applied after the dry/wet blend. Use to compensate for the level reduction that results from a narrow bandwidth curve or a sparse modulator.

3.2 Vocoder Display

The large panel below the knobs is the **vocoder display**. It always shows the live band-level bars, and overlays one of two editable curves depending on the active editing mode (toggled by the button in the top-left corner of the display):

- **Band-level bars**: real-time amplitude of each vocoder band, drawn as vertical bars. The height represents the envelope value for that band normalised and clamped to the display height. The sand-coloured cap at the top of each bar marks the peak.
- **Bandwidth curve mode (BANDWIDTH button)**: the per-band filter width setting. The vertical position of each dot controls the Q factor of that band's filter pair. High position = wide band (low Q, blending into neighbours). Low position = narrow band (high Q, more precise but quieter).
- **Band Volume curve mode (BAND VOL button)**: a per-band output level multiplier in the range 0.0–1.2×. The curve is painted as a bright line; a dashed reference line marks the 1.0× (unity) level. Dragging the curve down silences individual bands; dragging it above the dashed line boosts them up to 1.2×.

Editing the Bandwidth Curve

Click anywhere in the display to set the bandwidth of the band under the cursor. Drag horizontally to paint a curve across multiple bands. The cursor changes to a crosshair when hovering over the display. The bandwidth curve is saved with the preset. It survives changing the band count: existing values are kept and new bands (when increasing count) are interpolated.

Editing the Band Volume Curve

Click the **BANDWIDTH** button in the top-left corner of the display to switch to **BAND VOL** mode. Click again to return to bandwidth editing. In Band Vol mode, click or drag anywhere in the display to draw a per-band output level multiplier. The top of the display represents 1.2× (maximum boost); the bottom represents 0.0× (full mute). The dashed horizontal line marks 1.0× (unity, no change).

The Band Volume curve multiplier is applied in the synthesis stage, after envelope following, so it acts as a spectral gain shaper on the vocoded output rather than on the analysis path. Muting a band with the Band Volume curve removes it from the output entirely; its envelope follower still tracks the modulator, so restoring the level brings the band back in immediately at the correct gain.

The Band Volume curve is saved with the preset alongside the Bandwidth curve and all knob values. The editing mode (BANDWIDTH or BAND VOL) is also remembered across sessions.

4. Presets and State

VOCODE uses the standard JUCE AudioProcessorValueTreeState mechanism. All knob values, the full 256-point bandwidth curve, and the full 256-point band volume curve are saved and restored by the host as part of the plug-in state. The bandwidth curve is stored under the XML key **BandwidthCurve** and the band volume curve under **BandVolumeCurve**, both as child elements of the XML state tree. Presets that pre-date the Band Volume feature load with all band volumes at 1.0× (unity).

5. CPU Usage Guide

The processing cost of VOCODE scales with:

- **BANDS × BAND ORDER:** each band runs 1–4 biquad filters on both modulator and carrier channels per sample. 256 bands at order 4 = 3072 multiply-accumulate operations per sample per channel.
- **Sample rate:** higher sample rates proportionally increase cost because more samples are processed per second.
- **Buffer size:** the filter-rebuild triggered by a band-count change happens once per block and is not sample-accurate, but it is fast and should not cause glitches except at very small buffer sizes (< 32 samples).

Approximate CPU cost on a modern desktop CPU at 44.1 kHz / 512-sample buffer:

BANDS	ORDER 1	ORDER 2	ORDER 4
16	< 1%	< 1%	< 1%
64	~1%	~2%	~4%
128	~2%	~4%	~8%
256	~4%	~8%	~16%

6. Tips and Techniques

Bandwidth Curve Shapes

The bandwidth curve is essentially a per-band equaliser for the filter Q. Some useful shapes:

- **Flat (mid):** neutral starting point. All bands equally wide.
- **Ramp low-to-high:** wide low bands, narrow high bands. Keeps bass warmth while sharpening high-frequency detail.

- **Inverted arch:** narrow at extremes, wide in the middle. Useful for reducing low-end mud and top-end hiss simultaneously.
- **Random jagged:** paint an irregular curve. Creates a comb-like frequency response character - experimental and interesting with synthesised sources.

Band Volume Curve Techniques

The Band Volume curve acts as a spectral gain shaper on the vocoded output. Because it is independent of the Bandwidth curve, both can be sculpted simultaneously. Some useful approaches:

- **Spectral EQ:** draw a smooth low-shelf or high-shelf to roll off muddy low-mids or tame harsh highs without touching the bandwidth settings.
- **Band muting:** drag isolated frequency regions to 0.0× to remove specific problem frequencies (e.g. a resonant room mode or a sibilant peak) from the vocoded output entirely.
- **Harmonic boost:** raise the curve to 1.2× at known harmonic frequencies of the carrier to emphasise the tonal character of the source.
- **Combined sculpting:** use narrow Bandwidth for precise analysis, then compensate for the resulting quiet bands by boosting those same bands in the Band Volume curve - fine spectral detail without the usual level penalty of high-Q filters.

7. Technical Reference

Filter Design

Each band uses a **transposed direct-form II biquad band-pass filter**. The band-pass topology is derived from the Audio EQ Cookbook formula. Q is computed once per block as:

```
baseQ = max(1, numBands * 0.35)    globalQ = baseQ * 2
perBandQ = globalQ * qMult(bandwidthCurve[b])
```

where `qMult` maps the bandwidth curve position [0, 1] to a Q multiplier [2.0, 0.5] - high curve position = lower Q = wider band, low position = higher Q = narrower band.

Envelope Follower

Each band has an independent first-order peak follower using an asymmetric AR model:

```
env = (|x| > env) ? att*env + (1-att)*|x| : rel*env + (1-rel)*|x|
```

Attack and release coefficients are computed as $\exp(-1 / (ms \times sr))$ where `sr` is the sample rate. This gives time-constant behaviour independent of sample rate.

Level Compensation

Without compensation, narrower bands (higher Q) carry less energy, and adding more biquad stages further reduces passband level. Three factors are pre-multiplied per band to maintain consistent loudness across different settings:

- **sqrt(qMult):** compensates for the $\sqrt{BW}$ energy reduction of narrower Q values.
- **sqrt(bandOrder):** compensates for passband narrowing when cascading multiple stages.
- **Spectral tilt $(fc / 1\text{kHz})^{0.30}$:** compensates for the natural ≥ 6 dB/octave energy roll-off in music and speech. Low bands are attenuated, high bands boosted, so the vocoded output has a balanced spectral weight.

Band Frequency Spacing

Bands are spaced **logarithmically** between 80 Hz and 12 kHz:

$$fc[b] = 80 \times \exp(\log(12000/80) \times b / (N-1))$$

This mirrors the logarithmic frequency resolution of the human auditory system, so each band covers a perceptually equal slice of the spectrum.

9. Credits and License

VOCODE is built with the JUCE framework (<https://juce.com>) with the help of Claude.ai by Anthropic.

The biquad coefficient formulas follow the Audio EQ Cookbook by Robert Bristow-Johnson, which is in the public domain.

This manual and all accompanying source code are provided "*as is*" without warranty of any kind.